



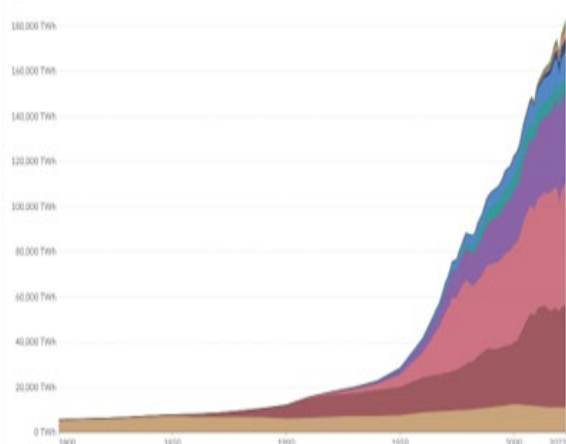
Horizon Kinetics Energy and Remediation ETF (NVIR)

Semi-Annual Report 2024 Commentary

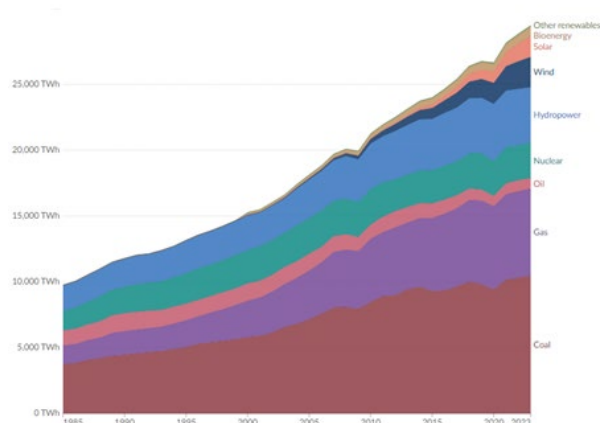
The objective of this fund is to focus on facts, not narrative. And in the world of energy, there is usually a vast difference between the two.

For example, worldwide demand for energy continues to increase, and the vast majority of energy produced in the world is derived from hydrocarbons, also known as fossil fuels. This percentage has declined only slightly in recent decades—from 76.8% in 2000 to 76.5% in 2023—despite trillions of dollars invested in wind and solar over that span. Even in terms of electricity generation, hydrocarbons accounted for 60.7% of worldwide production last year, only a modest decrease from 64.7% in 2000.

Global primary energy consumption by source
Primary energy is based on the substitution method and measured in terawatt-hours.



Electricity production by source, World
Measured in terawatt-hours.



Hydrocarbons remain popular because they are relatively easy to transport from the place of extraction to the place they are needed, and this can be done essentially without any loss of energy. That is not the case with electricity transportation or hydrogen transportation, which lose substantial quantities in transport. Hydrocarbons are also dense and relatively easy to store, and there are no losses of energy during the storage period, unlike some renewable sources. Hydrocarbons generally provide energy that is always on, unlike wind and solar, which are intermittent by definition—that is, dependent on weather to produce energy. Finally, hydrocarbons are relatively inexpensive to extract, refine, and convert into energy.

That is why efforts to reduce hydrocarbon use, led mostly by the U.S. and Europe, have not been successful on a grand scale. This is even true with coal, the primary target of such regulations, which reached a new high in terms of worldwide energy production last year, up 66% from 2000. For the reasons mentioned above, it is very difficult to displace hydrocarbons as the main source of global energy production.

Wind and solar are presently not adequate substitutes because of issues related to reliability (intermittency), transportation and storage, and cost. Nuclear energy, which is devoid of carbon dioxide emissions—and, for that reason, could be considered a better energy source from a climate change perspective—has declined lately. Nuclear accounted for the same amount of energy last year as it did in 1996, despite a 58.2% increase in total energy production since then.



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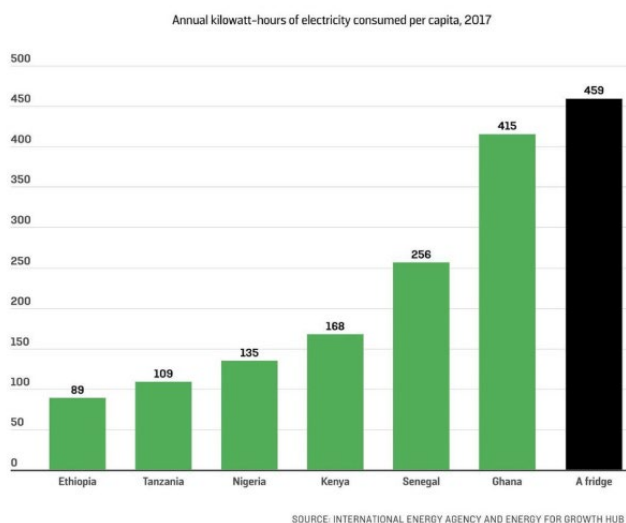
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Therefore, because the world's population needs energy to survive, hydrocarbons—rather than wind and solar—will almost certainly continue to dominate the energy mix for the foreseeable future. That is perhaps not the consensus view, but it is the view supported by the provided data. Therefore, rather than trying to replace hydrocarbons with wind and solar, producing hydrocarbons in the most environmentally friendly manner should become the new priority. Hence the “remediation” portion of the portfolio.

Should the energy production of the world continue to expand at same rate it has since 1996, which equates to 1.7% annualized, 289,869 Terawatt hours (TWh) will be produced in the year 2050. While there are talks about increasing nuclear production, there appears to be resistance toward it on multiple fronts—political, environmentalist, and public. Even if a substantial buildout of nuclear power were approved by regulators today, the lead time for planning, permitting, licensing, building, developing, and testing before a new nuclear reactor can come online is frequently 15-20 years. There is presently just one nuclear reactor under construction in the U.S.; 94 are in service.¹

On the demand side, there are several potential drivers that could result in dramatic increases in energy demand over the next few decades. While the Western-centric view appears to be that we should preserve energy as much as possible and try to limit—or even eliminate—the use of hydrocarbons, this is not a view shared with most of the people on the planet. Half the global population lives on less than \$6.85 per day, according to World Bank,² and their energy usage is essentially zero compared to that of the average American. In fact, many articles have been written about how the refrigerator in the average U.S. household consumes as much electricity as the entire daily energy use of the average person in many developing countries. For example, the chart below shows that in some countries in Africa, the energy consumption per capita is just 20%-25% of a typical U.S. household refrigerator.

Energy Use Per Person in Africa vs. a Typical American Refrigerator



The same is true for countries in southeast Asia, and the consensus seems to be that one average U.S. household refrigerator uses as much energy as the average human among the world's poorest 3 billion people. This does not appear to be very fair or equitable—or sustainable, for that matter. If the people of Africa were to meaningfully raise their standard of living over the next two to three decades, worldwide energy demand could increase dramatically.

For example, should the continent reach Mexico's 2023 per-capita energy consumption of 18,279 kWh³—compared to around 1,000 kWh now—Africa's power demand would increase not just 1,728%, but likely more. That's because its population is projected to

¹ <https://www.nei.org/resources/fact-sheets/u-s-nuclear-plants>

² <https://blogs.worldbank.org/en/developmenttalk/half-global-population-lives-less-us685-person-day>

³ <https://ourworldindata.org/grapher/per-capita-energy-use>



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expand by around 50% over the next two or three decades. This alone would increase worldwide energy consumption by 30%.

Another major source of rising demand is technology. Electric vehicles, 5G cellular services, quantum computing, artificial intelligence, and “the cloud”—in other words, enormous data centers—account for an ever-increasing share of global demand. This appears to be accelerating. For example, there are more than 5,500 data centers in the U.S., but only 448 in China.⁴ Given the strategic importance of AI and the country’s considerably greater population, it appears highly likely that China will catch up in the number of data centers. This will require enormous quantities of power, most likely natural gas or coal-generated.

The greatest concentration of U.S. data centers is in the greater Washington, D.C. area, but utilities are struggling to keep up with demand. Electric power generation in the U.S. has been stagnant since 2005, and the enormous push to place all kinds of data in the cloud over the past two decades has resulted in rapid growth in the number of data centers. The demand for cloud computing is boosted by new generative AI capabilities delivered by the three largest cloud infrastructure companies: Amazon, Microsoft, and Google. According to Canalys,⁵ global cloud spending grew by 21% year-over-year in the first quarter of 2024. This is a notable acceleration compared to a 19% year-over-year growth demonstrated by the industry in the same period last year.

An emerging narrative is that AI technology is the investment theme of the future, whereas energy is one of the past. In reality, the two are closely symbiotic. For example, energy companies need technology to improve their exploration, production, and transportation activities. Technology companies, for their part, need access to substantial amounts of energy.

Nvidia expects to have sold 3.5 million of its H100 Graphic Processing Units (GPUs) by the end of 2024. These are referred to as AI GPUs and each one of them consumes 700 watts of power. This is equivalent to 6.1 Gigawatt hours (GWh) if operated 24/7 for one year.⁶ Hence, the 3.5 million H100 chips would consume 21.5 Terawatt hours (TWh) in aggregate. That exceeds the annual power consumption of nations like Georgia, Lithuania, or Guatemala.

Since 2005, the U.S. has produced roughly 4,000-4,200 TWh of electricity every year. Thus, the 3.5 million Nvidia AI GPUs would represent around 0.5% of the electricity production in the U.S. But Nvidia just started selling the H100 AI GPU, so all that electricity demand will be incremental, and likely to be increasing rapidly over the next few years. The H100 has been shipping for just about one year, replacing the A100 chip, which consumed 400W. The recently announced Blackwell B200 chip, which is scheduled to enter production next year, is expected to consume 1,000W. Thus, with every upgrade cycle, there could be millions of incremental units sold, with each requiring more power than the previous generation.

The CEO of fuel cell maker Bloom Energy has stated that “AI-powered search on a GPU will take 5 to 20 times more power per search compared to a regular search on a CPU, and the use of AI is growing rapidly across all segments of our society.”

⁴ <https://www.statista.com/statistics/1415287/apac-data-center-number-by-country/>

⁵ <https://www.canalys.com/newsroom/global-cloud-market-Q1-2024>

⁶ <https://www.tomshardware.com/tech-industry/nvidias-h100-gpus-will-consume-more-power-than-some-countries-each-gpu-consumes-700w-of-power-35-million-are-expected-to-be-sold-in-the-coming-year>



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If this AI-related upgrade from CPUs to GPUs fully materializes,⁷ and if data centers currently account for 3% of the world's electricity demand,⁸ they would soon require copious amounts of incremental power. This would almost certainly result in considerably higher electricity prices across the grid due to limited supply and spiking demand. But the incremental electricity needs to be generated first, otherwise it simply will not be available—at any price.

Perhaps even more problematic: If the EV transition is successful—and 10-15 years from now every vehicle in the U.S. is electric, not powered by an internal combustion engine—a ballpark calculation indicates that electricity needs would more than double. Data centers would then have to compete with the EV charging market for what is likely much more expensive and much less available electricity.

As a result, energy companies may also be revalued, but to considerably higher levels, as their products become scarcer (and therefore, valuable). Their services are set to become more important in keeping the growth in the technology sector going, and to position the U.S. well in the global race for AI supremacy.

Of course, this energy should be generated in the most environmentally friendly manner possible, which explains why the Horizon Kinetics Energy and Remediation ETF has a dual mandate. We invest in both energy companies and the environmental remediation companies with solutions to allow energy to be extracted, processed, and dispatched with as little impact as possible on the environment.

In terms of portfolio developments, ExxonMobil was the largest position in the fund at the end of the second quarter, at 7.9%. While the fund did not invest in Exxon initially, the company became the largest position by first acquiring Denbury Inc. in the fourth quarter of 2023, and then Pioneer Natural Resources in the second quarter of 2024, both of which were held in our portfolio.

The Denbury acquisition highlights the value perceived by Exxon in carbon capture, utilization, and storage (CCS) solutions, while the Pioneer acquisition doubled Exxon's footprint in the Permian Basin. Also, using combined operating capabilities and infrastructure, Exxon expects to increase the amount of recycled water used in its Permian fracturing operations to more than 90% by 2030 as a result of the Pioneer acquisition.

Consequently, these two acquisitions by Exxon can be considered examples of the two investment qualities this fund seeks to obtain exposure to: increasing energy production, and doing so in an environmentally responsible manner.

⁷ That is, all of the existing GPUs will be upgraded with the newer AI GPUs.

⁸ Based on data from the International Energy Agency.



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Horizon Kinetics Energy and Remediation ETF Top 10 Holdings (%) as of June 30, 2024

Exxon Mobil Corp	7.9%
Diamondback Energy Inc	4.8%
Williams Cos Inc/The	4.4%
CES Energy Solutions Corp	3.9%
Cheniere Energy Inc	3.9%
EQT Corp	3.9%
Texas Pacific Land Corp	3.7%
PrairieSky Royalty Ltd	3.7%
Suncor Energy Inc	3.7%
ConocoPhillips	3.5%

IMPORTANT RISK DISCLOSURES

Please consider carefully a fund's investment objectives, risks, charges and expenses. For this and other important information, obtain a statutory prospectus and summary prospectus by contacting 646-495-7333. Read it carefully before investing.

The Horizon Kinetics Energy and Remediation ETF (Symbol: NVIR) is an exchange traded fund managed by Horizon Kinetics Asset Management LLC ("HKAM").

Investing involves risk, including the possible loss of principal. Shares of any ETF are bought and sold at market price (not NAV), may trade at a discount or premium to NAV and are not individually redeemed from the Fund. Brokerage commissions will reduce returns. The Fund's investments in securities linked to real assets involve significant risks, including financial, operating, and competitive risks. Investments in securities linked to real assets expose the Fund to potentially adverse macroeconomic conditions, such as a rise in interest rates or a downturn in the economy in which the asset is located.

The Fund is non-diversified, meaning it may concentrate its assets in fewer individual holdings than a diversified fund. Therefore, the Fund is more exposed to individual stock volatility than a diversified fund.

Fund holdings and sector allocations are subject to change at any time and should not be considered recommendations to buy or sell any security.

The Fund invests in foreign securities which involve greater volatility and political, economic and currency risks and differences in accounting methods. These risks are greater for investments in emerging markets.



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The Fund may invest in the securities of smaller and mid-capitalization companies, which may be more volatile than funds that invest in larger, more established companies. The fund is actively managed and may be affected by the investment adviser's security selections.

Diversification does not assure a profit or protect against a loss in a declining market.

Energy Sector Risk: The Fund's investments are exposed to issuers conducting business in the Energy Sector, including energy, industrial, infrastructure, and logistics companies, and is therefore susceptible to the adverse economic, environmental, business, regulatory, or other occurrences affecting the Energy Sector. Companies in the Crude Petroleum and Natural Gas Industry are affected by specific risks, including, among others, fluctuations in commodity prices; reduced consumer demand for commodities such as oil, natural gas, or petroleum products; reduced availability of natural gas or other commodities for transporting, processing, storing, or delivering; slowdowns in new construction; extreme weather or other natural disasters; and threats of attack by terrorists on energy assets. Additionally, Crude Petroleum and Natural Gas Industry companies are subject to substantial government regulation and changes in the regulatory environment for energy companies may adversely impact their profitability.

HKAM does not provide tax or legal advice, all investors are encouraged to consult their tax and legal advisors regarding an investment in the Fund. You may obtain additional information about HKAM at our website at www.horizonkinetics.com.

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